

PHD21 Computation methods: Replication report

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I have picked two models which constitute the building blocks of my first-year project. In this context, I put more weight on the intensive rather than the extensive margin of the exercise. I use Julia and teach myself all the key “tricks” to speed up the computation process, prioritising precision and replicability. In the case of Buera et al. (2015), who provide no replication package and give scant details on the values of the model’s moments, I remain as faithful to the author’s described methods. In so doing, I like to think that I conduct a “stress test”, which I believe the paper does not fully pass. When it comes to Elsby and Michaels (2013), which is more widely used and replicated, I translate the model to Julia and use the repeated transition method of Lee (2025) to improve on the authors’ current methodology (Krusell and Smith, 1998).

Note. The code extensively uses Julia’s functionality by allowing for a wide range of UNICODE characters, such as Greek letters and vector symbols. My [GitHub page](#) provides further instructions on how to install the special font need to use these extra characters. Without doing so, the code may be quite difficult to read.

1 Elsby and Michaels (2013): Marginal jobs, heterogenous firms

1.1 Overall impressions

Elsby and Michaels (2013)’s model is quite elegant and parsimonious, as is the case with its original algorithm. Their MATLAB version is quite slow, so I have taken it to Julia to speed it up, on top of making it a bit more concise. This has remained a challenge, as the authors apply many tricks to tackle the non-linearity around the hiring and firing decisions. Specifically, they update the labour grid every period, which yields a very clean solution, though it throws sand in the gears of Julia’s engine. This is especially the case in the RTM adaptation of the model, which gets quite slow (still very fun, though).

1.2 Framework

In the model, there’s a unit mass of firms which are heterogeneous along their idiosyncratic productivity (x) dimension. They hire n workers, which is subject to search and matching frictions. The authors adapt Stole and Zwiebel (1996)’s bargaining protocol to a setting with diminishing returns.

With the hiring costs sunk at the time of negotiations, the marginal surplus is given by:

$$(1) \quad J(n, x) = p x F'(n) - w(n, x) - w_n(n, x) n + \beta D(n, x),$$

where the last term represents the marginal effect of current employment decisions on the firm's value. The firm's decision is to choose the number of workers given x and the previous period's number of employees.

$$(2) \quad n(n_{-1}, x) = \begin{cases} R_v^{-1}(x) & \text{if } x > R_v(n_{-1}) \\ n_{-1} & \text{if } x \in [R(n_{-1}), R_v(n_{-1})] \\ R^{-1}(x) & \text{if } x < R(n_{-1}), \end{cases}$$

where $R_v(n_{-1})$ and $R(n_{-1})$ are hiring and firing policies. $n(n_{-1}, x)$'s piece-wise nature is what makes the computation a challenge. The rest of the model is standard.

1.3 Steady state

Algorithm 1 outlines how the steady state equilibrium is calculated. The repeated cubic interpolations (necessitated by the ever-adjusting labour grid, $\mathcal{N}$) are the algorithm's main bottleneck. Still, the convergence comes quickly, after around 3–8 seconds. Table 1 compares my replication with the results in Elsby and Michaels (2013).

Table 1: Steady state equilibrium

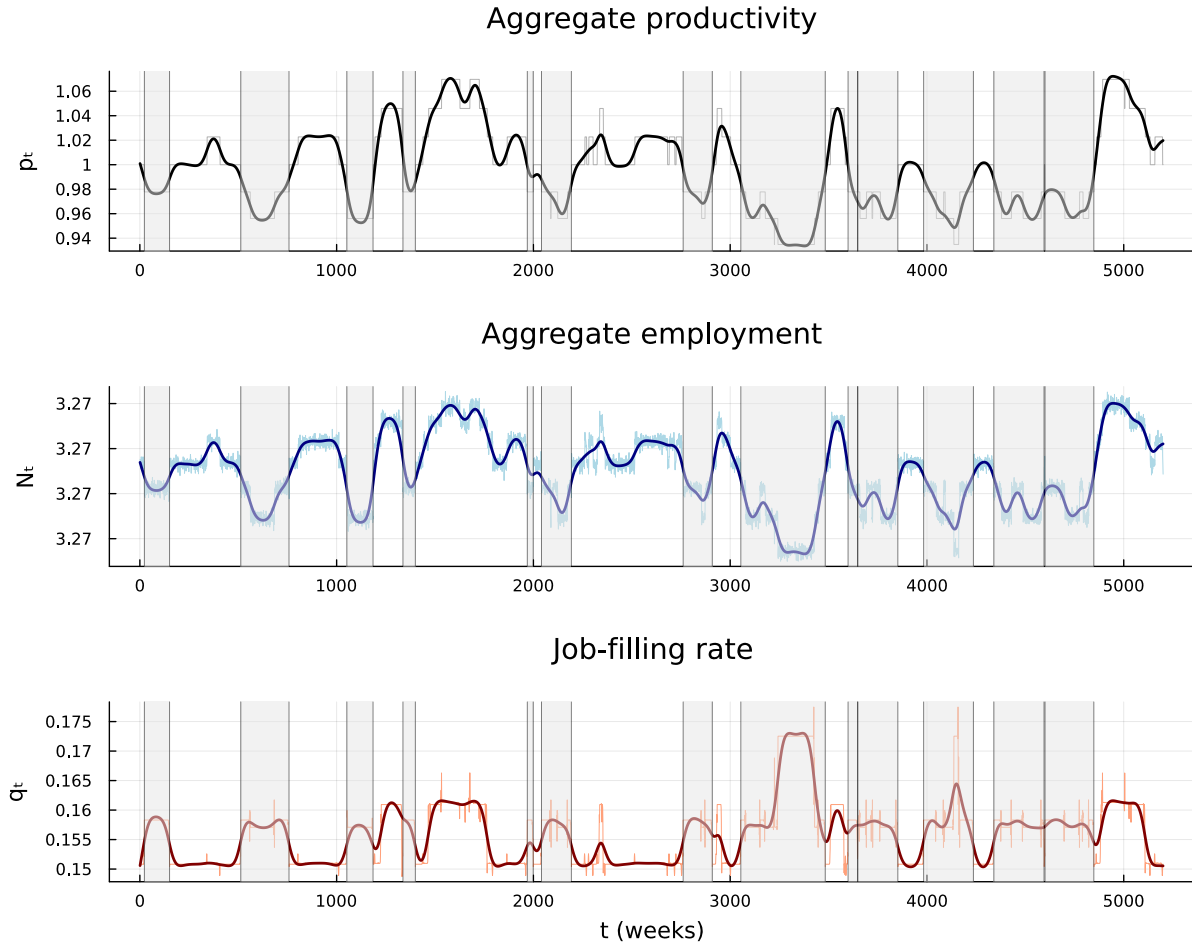
	Data	EM (2013)	Replication
Job-finding rate, f	0.1125	0.1125	0.1112
Inflow rate, s	0.0078	0.0078	0.0073
Unemployment rate, U/L	0.0650	0.0650	0.0658
Vacancy-unemp. ratio, θ	0.7200	0.7200	0.6934
Vacancy rate, V/L	—	—	0.0456

1.4 Aggregate uncertainty: Repeated transition method

Algorithm 2 outlines the application of the RTM to this model (Lee, 2025). The model performs quite well in the RTM environment, but given its design, it's very slow. What I'm presenting below is the best result coming from a week of experimentation.

Figure 1 illustrates the behaviour of the key macro aggregates. The top plot shows the simulated path of productivity. The productivity process is the same as the one used by Elsby and Michaels (2013). For exposition purposes, I smooth it using the HP filter, picking the smoothing parameter from the original paper. I also shade “recessions” — the periods when p_t has been less than the mean for at least two quarters. This allows for distinguishing the variables' behaviour in booms and recessions. The remaining two panels flesh out the dynamics of the key macroeconomic aggregates. When it comes to the aggregate employment, the direction of travel is satisfactory, but not volatile enough. This is entirely attributable to the very small updating weight I've been using in the updating procedure. The story is similar, albeit a bit better, for the job-filling rate. In what will follow beyond the deadline, I need to tinker with the updating weights for the key variables.

Figure 1: Simulated and estimated variable paths



Notes: The figure plots simulated paths of aggregate productivity p_t , employment N_t , and the job-filling rate q_t from the RTM solution (Lee, 2025). The thin lines denote weekly values while the bold lines denote the HP-filtered trend with smoothing parameter $\lambda = 100,000$, which matches Elsby and Michaels (2013)'s approach. The shaded regions indicate episodes in which aggregate productivity remains below its steady-state value for at least two consecutive quarters, 24 weeks (take that, NBER).

Table 2 compares the key variables' estimates of the elasticity with respect to the productivity. I'm still way off, which is mostly attributable to the low volatility of the aggregate employment and job-filling rate. With the low updating weights, the model stagnated in a very "stable" place.

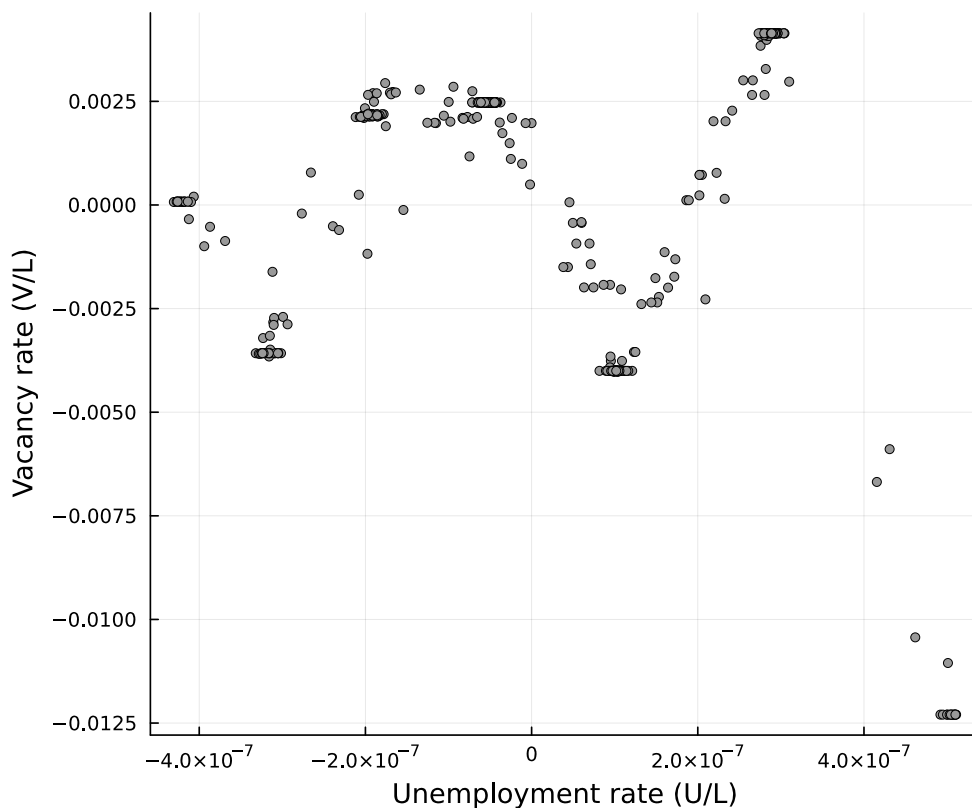
Table 2: Elasticities

	Data	EM (2013)	Replication
Job-finding rate, f	2.65	2.55	0.25
Inflow rate, s	-1.89	-1.64	2.72
Vacancies, V	2.91	2.47	3.10
Tightness, $\theta = V/U$	6.44	6.37	3.10

Figure 2 reproduces Elsby and Michaels (2013)'s Beveridge curve. Believe it or not, as I have seen multiple runs of the model, this is actually getting closer to what the original version shows. To improve it, I need to find a right set of updating weights that do not make the model

stagnate and do not make it explode (I genuinely think it's not an empty set).

Figure 2: Beveridge curve



Notes: Variables are demeaned to match the construction of Figure 5 in Elsby and Michaels (2013).

2 Buera et al. (2015): Anatomy of a credit shock

2.1 Overall impression

Having initially treated the paper as a potential “easy win,” I have been humbled by the model’s fragility. The authors’ results cannot be fully replicated, which I believe to be mostly attributable to the framework being extremely sensitive to minor changes to the productivity process. I also suspect that Buera et al. (2015) use more tricks to find the MIT shock paths reported in the paper, especially given that the described procedure is a baseline approach to solve the Aiyagari (1994) class of models. Needless to say, I have learnt a lot while working on this model.

2.2 Framework

The model features agents who are heterogeneous along two dimensions, their wealth (a) and entrepreneurial productivity (z). If an agent chooses to be an entrepreneur, she hires l workers. If she remains a worker, she earns wage w_t , irrespective of her employment status. Workers can be unemployed, with the job destruction endogenously determined by the downsizing entrepreneurs and the job finding rate, the latter of which is determined by the number of

workers and job openings.

The value function for agent $i \in \{e, w\}$ is:

$$(3) \quad \begin{aligned} V^i(a, z) = & \max_{a', c} \{u(c) + \beta \mathbb{E}[V(a', z)]\} \\ \text{s.t } & c + a' = (1 + r)a + \Pi(a, z)\mathbb{1}\{i = e\} + w_t\mathbb{1}\{i = w\} - \tau, \end{aligned}$$

where $u(c)$ is the utility of consumption c , β is the discount rate, r is the interest rate, $\Pi(a, z)$ is the profit level, and τ is the tax financing the unemployment benefit (which is set to be the same as the wage).

Entrepreneur (a, z) earns the following profit:

$$(4) \quad \begin{aligned} \Pi(a, z) = & \max_{k, l} \{Azk^\alpha l^\theta - wl - (r + \delta)k\} \\ \text{s.t } & k = \max\{k^*, \lambda a\}, \end{aligned}$$

where λ introduces the collateral constraint.

2.3 Steady state

Algorithm 3 fleshes out Buera et al. (2015)’s steady state solution approach. While pedagogically clean, the method is relatively inefficient – I keep using it to remain faithful to the paper (which does not have a replication package). Table 3 summarises how my replication fares against the authors’ values. While I manage to match the top 10% earnings share, the establishment exit rate, and the credit market instruments to non-financial assets, I struggle with the real interest rate and the top 10% employment share.

Table 3: Comparing data, original model, and replication

Moment	US data	Original model	Replication
Top 10% employment	0.69	0.69	0.88
Top 5% earnings share	0.30	0.30	0.32
Establishment exit rate (annual)	0.10	0.10	0.11
Real interest rate (annual)	0.0200	0.0200	0.0070
Credit market instruments to non-financial assets	0.7000	0.7000	0.7110

In my experiments, I find that this is likely attributable to the construction of the productivity grid. Buera et al. (2015) discuss the construction process in the paper’s appendix, where they outline the manual procedure employed to construct the Pareto grid. If they instead use a standard package that treats the grid’s last two points even slightly more differently, the two mismatched moments behave slightly better (still not perfectly in line with the Table 3).¹ The idea is that even small adjustments to the grid lead to a smaller average size of the most productive establishments. This, in turn, leads to lower savings of their owners, pushing up the real interest rate. If I “naively” run the model with 40 – 80 wealth grid points or with a very

¹I have tried and failed to backward-engineer an “ideal” productivity process that would simultaneously match the moments and instructions. The only time when I got closer was when I significantly departed from their procedure.

restrictive value for the maximum wealth, the result also gets closer to that of Buera et al. (2015).

To further test the model’s performance, I compare and contrast two alternative methods of solving the value function iteration routine. First, I search for a solution on the grid, which is set to have 40 and 400 points for productivity and assets respectively.² This is fast, but potentially less accurate, with the VFI routine frequently stalling for lower asset grid sizes. Second, I search for a solution using the interpolated values (for both V^e and V^w to avoid smoothing over the kink). The values are essentially the same.

2.4 Aggregate dynamics: A MIT shock

Algorithm 4 sets out the method for computing the MIT shock-induced transition dynamics. The algorithm consists of three nested loops, which I have found to be quite sensitive. Following Buera et al. (2015), I set the number of periods to 70 (like them, I have also tested the model for 100 periods) and a path for the borrowing constraint shock. In what follows, I argue that a solution matching the paper’s results is near-impossible to find, which I believe to stem from the algorithm’s construction.

Specifically, for price $x \in \{w, \tau, r\}$, they find x^{implied} – the value clearing the market at the current distribution, which is used to update the next guess. Consider the $r^{(n)}$ guess. Given $r^{(n)}$, we compute the distribution, $G^{*(n)}(\cdot)$, and the prices, $\tau^{*(n)}$ and $w^{*(n)}$. From there, we get $r^{\text{implied},(n)}$, which informs $r^{(n+1)}$ via convex updating. The real interest rate, like all prices, informs the occupational policy, which may jump in certain areas of r ’s support, making the result jittery. This is precisely what is happening in Figure 3, which shows the convergence pattern for r . It’s worth bearing in mind that this problem happens for all variables, not just r , making the solution rather unstable. A few alternative solutions are possible:

1. an enormous asset grid ($N_a > 1000$)
2. smart interpolations combined with letting the code run for hours or days³
3. the endogenous grid method (which is what the authors definitely didn’t do according to the paper)
4. another solver (at least for w and τ)
5. non-convex updating⁴
6. adapting the grid during every iteration to capture the area around the kink.

²Buera et al. (2015) only specify the size of the productivity grid, which I keep unchanged in all experiments. The wealth grid does not matter as much as I expected - the presented steady state result remains relatively similar for all values of $N_a \geq 100$.

³Given time constraints, I opted against using cubic interpolations while solving this model. The steady state equilibrium is the same whether I solve it on the grid or interpolate everything that moves. At the same time, using any interpolation method massively increases the time needed to solve the model. Another common problem with non-linear interpolation is that extra care needs to be exercised to avoid “unreasonable” numbers.

⁴This is what I did in one experiment a few weeks ago. It didn’t move things much further, but it was generally more well-behaved.

Figure 3: MIT shock convergence dashboard

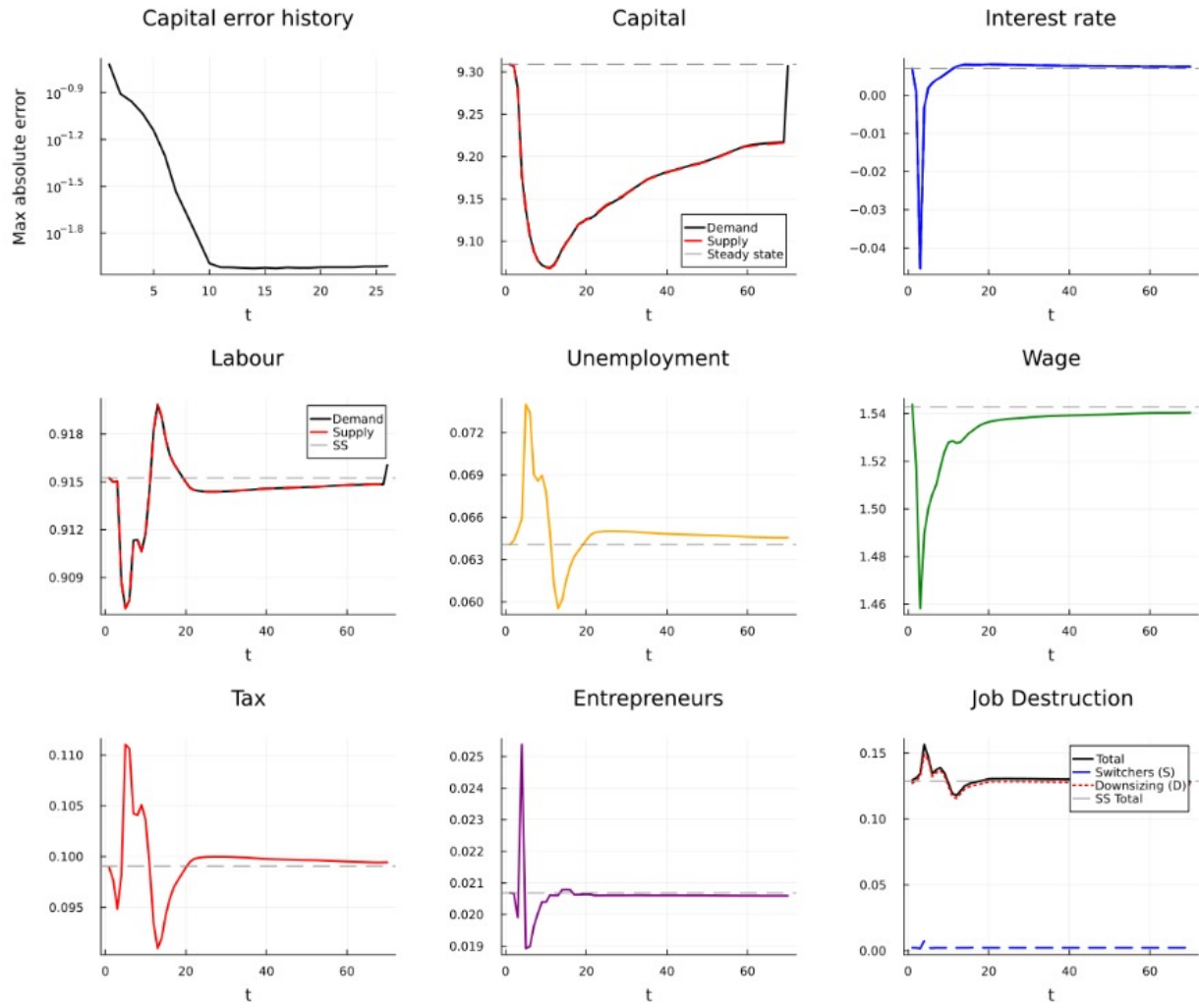


Figure 4: MIT shock transitional dynamics

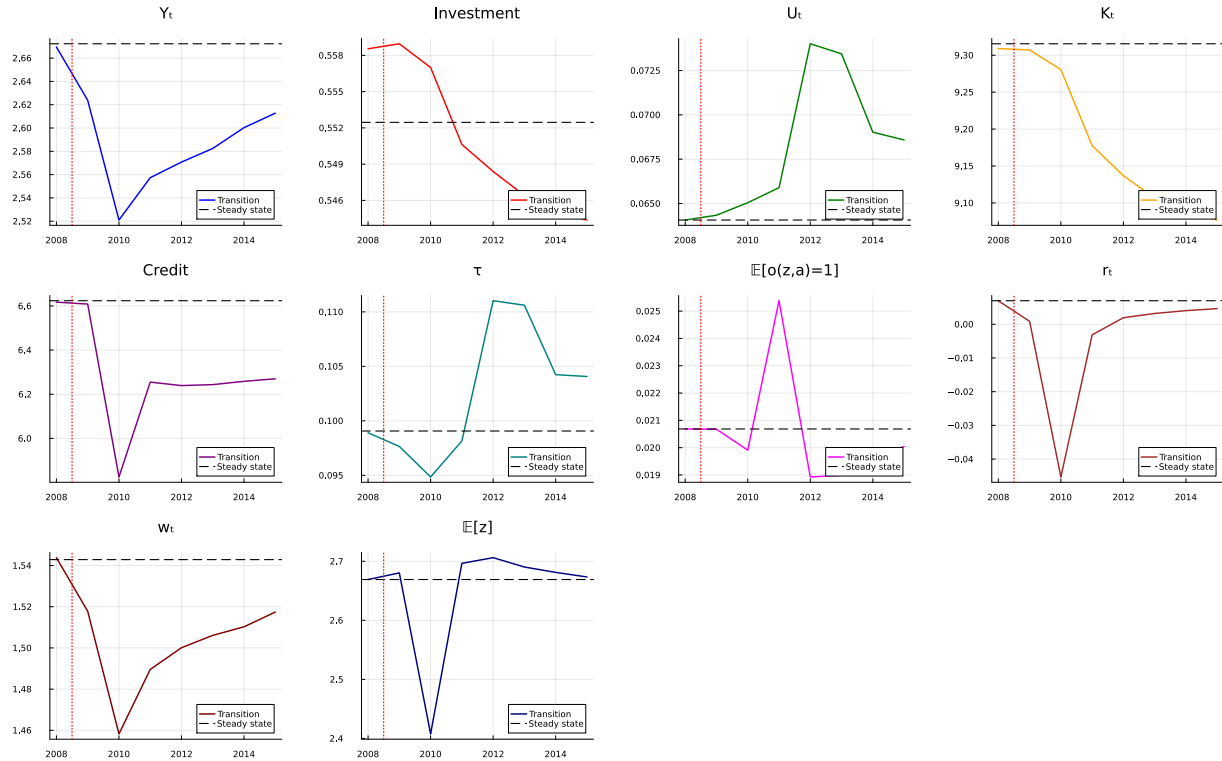


Figure 4 replicates Figures 6 and 7 from Buera et al. (2015)’s work. What honestly surprises me that this less than perfect solution is not far when it comes to the output, TFP, unemployment, wages, and interest rate. The main difference lies in the investment rate, which is considerably off (I find the shock’s persistence much higher). For a solution that works mostly thanks to the very generous exit conditions and my weariness, this isn’t bad.

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A Elsby and Michaels (2013)’s algorithms

Algorithm 1: Stationary equilibrium (Elsby and Michaels, 2013)

Input: Parameters $(\eta, \alpha, \beta, c, b, \mu, \varepsilon, \lambda, \xi)$, aggregate productivity p , grids $(\mathcal{X}, \mathcal{N})$

Output: Job-filling rate q^* , policies $(R(n), R_v(n))$, aggregates (N, Y, S, M)

// Outer loop: find q^* via bisection

1 **repeat**

2 Compute $f = \mu \left(\frac{q}{\mu} \right)^{(\varepsilon-1)/\varepsilon}$ from the matching function;

3 Construct endogenous grid $\mathcal{N}(p, f, q)$ and wage matrix $W(x, n)$;

 // Stage 1: Coarse VFI on initial grid

4 Initialize $\Pi^0(x, n)$ from scaled flow profits;

5 **repeat**

6 Compute $\Pi^c = \Pi^{\text{flow}} + \beta \mathbb{E}_x[\Pi^k]$;

7 Solve $n^f(x) = \arg \max_n \Pi^c(x, n)$ and $n^h(x) = \arg \max_n \Pi^c(x, n) - \frac{c}{q} n$;

8 Set $\Pi^{k+1}(x, n) = \max \{ \Pi^f(x, n), \Pi^h(x, n), \Pi^c(x, n) \}$;

9 **until** $\frac{\|\Pi^{k+1} - \Pi^k\|_\infty}{\|\Pi^k\|_\infty} < \delta^{\text{ref}}$;

 // Stage 2: Refined VFI on adaptive grid

10 Refine $\mathcal{N}$ around $[n_1^h, n_{N_x}^f]$; interpolate $\mathbb{E}_x[\Pi]$ onto new grid;

11 **for** $s = 1, \dots, \bar{n}^s$ **do**

12 Solve firing/hiring via (cubic) spline optimisation over n ;

13 Update Π^{k+1} as above;

14 **end**

 // Extract policies and aggregate

15 Invert $n^f(x) \rightarrow R(n)$ and $n^h(x) \rightarrow R_v(n)$ via spline interpolation;

16 Compute N, S , and U ;

17 Update q by bisection on residual $N^{\text{implied}} - N$;

18 **until** $|N^{\text{implied}} - N^{\text{aggregated}}| < \varepsilon_q$;

Algorithm 2: Application of Lee (2025)'s RTM to Elsby and Michaels (2013)'s model

Input: Parameters, shock path $\{p_t\}_{t=1}^T$, tolerance δ_{RTM} , relaxation weights $(\omega_1, \omega_2, \omega_3)$

Output: Equilibrium paths $\{q_t^*, N_t^*, \Pi_t^*, R_t^*, R_{vt}^*\}$

```
1 Initialise predicted paths  $\{q_t, N_t, \Pi_t\}_{t=1}^T$  from steady state or warm start;
2 repeat
    // A. Backward induction
3   for  $t = T$  to 1 do
4       Construct  $\mathbb{E}_t[\tilde{\Pi}_{t+1}]$  via interpolation matching counterfactual  $N$  and  $p_{t+1}$  states;
5       Solve Bellman equation for target firm value  $\Pi_t^{\text{imp}}$ ;
6       Extract firing threshold  $R_t$  and hiring threshold  $R_{vt}$ ;
7   end
    // B. Forward simulation station
8   Set  $N_0$  to initial steady-state employment;
9   for  $t = 1$  to  $T$  do
10      Compute aggregate separations  $S_t$  and matches  $M_t$  using thresholds  $(R_t, R_{vt})$ ;
11      Update employment:  $N_t^{\text{imp}} = N_{t-1} + M_t - S_t$ ;
12      Find implied job finding rate  $f_t$  and compute implied tightness  $q_t^{\text{imp}}$ ;
13      Clamp  $q_t^{\text{imp}}$  bounds and store implied paths;
14   end
    // C. Convergence and dampening
15    $\varepsilon_{RTM} \leftarrow \max_t |q_t - q_t^{\text{imp}}|$ ;
16   Update  $q_t \leftarrow \omega_1 q_t^{\text{imp}} + (1 - \omega_1) q_t$ ;
17   Update  $N_t \leftarrow \omega_2 N_t^{\text{imp}} + (1 - \omega_2) N_t$ ;
18   Update  $\Pi_t \leftarrow \omega_3 \Pi_t^{\text{imp}} + (1 - \omega_3) \Pi_t$ ;
19 until  $\varepsilon_{RTM} < \delta_{RTM}$ ;
```

B Buera et al. (2015) algorithms

Algorithm 3: Stationary equilibrium (Buera et al., 2015)

Input: Parameters $(\sigma, \beta, \alpha, \theta, \delta, \psi, \eta, \lambda)$, grids $(\mathcal{A}, \mathcal{Z}, \mathcal{L})$, tolerances $(\delta_r, \delta_\tau, \delta_w, \delta_v, \delta_d)$

Output: Prices (r^*, w^*, τ^*) , policies (c, a', k, l, o) , distribution G^* , unemployment U^*

```

1 Initialize  $r^{(0)}$ ;
2 repeat
3   Initialize  $\tau^{(0)}$ ;
4   repeat
5     Initialize  $w^{(0)}$ ;
6     repeat
7       // Value function iteration
8       repeat
9         Compute  $v^W(a, z)$  from Equation (3);
10        Compute  $v^E(a, z)$  from Equation (3) using static FOCs for  $(k, l)$  s.t.
11           $k \leq \lambda a$  from Equation (4);
12        Set  $v(a, z) = \max\{v^W, v^E\}$ ; record  $o(a, z), a'(a, z)$ ;
13      until  $\|V^{n+1} - V^n\|_\infty < \varepsilon_v$ ;
14      // Forward simulation
15      Iterate  $G_{n+1} = \mathcal{T}(G_n; a', o, \psi, \mu)$  until  $\|G_{n+1} - G_n\| < \varepsilon_G$ ;
16      Compute  $L^d = \int l dG, L^s = 1 - \int \mathbb{1}\{o = E\} dG - U$ ;
17      Update  $w$  by bisection;
18    until  $|L^d - L^s| < \varepsilon_w$ ;
19    Compute  $U$ , set  $\tau^{\text{new}} = wU$ ;
20    Update  $\tau \leftarrow \eta_\tau \tau^{\text{new}} + (1 - \eta_\tau)\tau$ ;
21  until  $|\tau^{\text{new}} - \tau| < \varepsilon_\tau$ ;
22  Compute  $K^d = \int k dG, K^s = \int a dG$ ;
23  Update  $r$ , raising if excess demand and lowering if excess supply. ;
24 until  $|K^d - K^s| < \varepsilon_r$ ;

```

Algorithm 4: MIT shock transition algorithm (Buera et al., 2015)

Input: Parameters, shock paths $\{A_t, \lambda_t\}_{t=1}^T$, steady state $\{V_{ss}, G_{ss}, \dots\}$.

Output: Price paths $\{r_t^*, w_t^*, \tau_t^*\}$, distributions G_t^* , aggregates

```
1 Initialise paths  $r_t^{(0)}, w_t^{(0)}, \tau_t^{(0)}$  for  $t \in 1, \dots, T$ ;
2 repeat
3   Initialise  $\tau_t^{(0)}$ ;
4   repeat
5     Initialise  $w_t^{(0)}$ ;
6     repeat
7       // A. Backward logic
8       Set  $V_T, o_T, k_T, a'_T, l_T$  to steady state;
9       for  $t = T - 1$  to 1 do
10        Compute expected value  $\mathbb{E}V_t$  using  $V_{t+1}$ ;
11        Solve static policies for workers and entrepreneurs;
12        Solve for asset choices  $a'_t$ , occupational choice  $o_t$ , and  $V_t$ ;
13      end
14      // B. Forward simulation
15      Set  $G_1 = G_{ss}$ ;
16      for  $t = 1$  to  $T - 1$  do
17        Compute job destruction and labour market aggregates;
18        Compute transition flows;
19        Update distribution  $G_{t+1}$ ;
20      end
21      // C1. Labour market residual
22      for  $t = 1$  to  $T$  do
23        Compute aggregates  $L_t^d$  and  $L_t^s$ ;
24        Find implied  $w_t$  by bisection to clear labour market;
25        Update  $w_t \leftarrow \eta_l w_t^{\text{implied}} + (1 - \eta_l) w_t$ ;
26      end
27      until  $\max_t |L_t^d - L_t^s| < \varepsilon_l$ ;
28      // C2. Government balance
29      Set  $\tau_t^{\text{new}} = w_t U_t$  for all  $t$ ;
30      Update  $\tau_t \leftarrow \eta_\tau \tau_t^{\text{new}} + (1 - \eta_\tau) \tau_t$ ;
31      until  $\max_t |\tau_t^{\text{new}} - \tau_t| < \varepsilon_\tau$ ;
32      // C3. Convex updating for MIT shock
33      for  $t = 1$  to  $T$  do
34        Compute aggregates  $K_t^d$  and  $K_t^s$ ;
35        Find implied  $r_t$  by bisection to clear capital market;
36        Update  $r_t \leftarrow \eta_c r_t^{\text{implied}} + (1 - \eta_c) r_t$ ;
37      end
38      until  $\max_t |K_t^d - K_t^s| < \varepsilon_r$ ;
```
